(†) The rings should be regarded as commutative rings with identity.

Definition 0.1. : Let $A \subseteq B$ be two rings, i.e., A is a subring of B. An element $x \in B$ is said to be **integral over** A if $\exists n \in \mathbb{N}$ and $\exists a_1, a_2, \ldots, a_n \in A$ such that

$$x^n + a_1 x^{n-1} + \cdots + a_{n-1}x + a_n = 0$$

Example 0.1. :Every element $x \in A$ is integral over A itself.

Example 0.2. : Let k be a field, $A = k[x^2], B = k[x]$. Then $x \in B$ is integral over A.

Example 0.3. : Let S_3 acts on $k[x_1, x_2, x_3]$:

$$S_3 \times k[x_1, x_2, x_3] \to k[x_1, x_2, x_3]$$

$$(\sigma, x_i) \mapsto \sigma(x_i) := x_{\sigma(i)}$$

i.e., for $p(x_1, x_2, x_3) \in k[x_1, x_2, x_3]$, we have $\sigma(p(x_1, x_2, x_3)) = p(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)})$.

Let $A := R^{S_3} \triangleq \{p \in k[x_1, x_2, x_3] \mid \sigma(p) = p\} = k[e_1, e_2, e_3]$, where $e_1 = x_1 + x_2 + x_3, e_2 = x_1x_2 + x_2x_3 + x_1x_3, e_3 = x_1x_2x_3$. Then $A \subseteq k[x_1, x_2, x_3]$. Consider the polynomial

$$p(y) := (y - x_1)(y - x_2)(y - x_3) = y^3 - e_1y^2 + e_2y - e_3$$

so we know that $p(x_i) = 0$ $(\forall i = 1, 2, 3)$ and hence x_i $(i = 1, 2, 3) \in k[x_1, x_2, x_3]$ are all integral over A.

Example 0.4. : Let R be a ring and G is a finite subgroup of $Aut(R)$. Let

$$A := R^G \triangleq \{r \in R |\ g(r) = r\ (\forall g \in G)\}$$

Now for $\forall r \in R$, take $P(y) = \prod_{g \in G}(y - g(r))$. Then $P(r) = 0$, which means every elemnet $r \in R$ is integral over A.

Let $A \subseteq B$ be two rings, consider

$$C := \{x \in B |\ x \text{ is integral over } A\}$$

then it's clear that $A \subseteq C$.

Theorem 0.1. *: Let $A \subseteq B$ be rings, then TFAE:*
(1) $x \in B$ is integral over A;
(2) $A[x]$ is a finitely generated A-module;
(3) there exists a finitely generated A-module C containing A and x;
(4) there exists a faithful $A[x]$-module M (i.e., $Ann(M) = \{0\}$) which is finitely generated as an A-module.

Proof: (1)⇒(2): Since $x \in B$ is integral over A, $\exists a_i \in A$ such that

$$x^n + a_1 x^{n-1} + \cdots + a_n = 0$$

thus $x^n = -a_1 x^{n-1} - a_2 x^{n-1} - \cdots - a_n$, which implies that $A[x]$ is generated by $\{1, x^1, x^2, \ldots, x^{n-1}\}$;

(2)⇒(3): Take $C = A[x]$;

(3)⇒(4): Take $M = C$, since $x \in C$ and $A \subseteq C$, so $A[x] \subseteq C = M$. Hence $1 \in A[x] = M$. Suppose $y \in A[x]$ such that $yM = 0$, therefore $y \cdot 1 = 0$, i.e., $y = 0$. Thus $Ann(M) = \{0\}$, M is a faithful $A[x]$-module and it's clear that M is finitely generated as an A-module;

(4)⇒(1): Consider the map

$$\varphi : M \to M$$

$$m \mapsto xm$$

so $\varphi \in End_A(M)$ and $\varphi(M) \subseteq M$. Suppose M is generated by $\{m_1, m_2, \ldots, m_n\}$ over A. Note that $AM = M$, by Caylay-Hamilton theorem, we can know that $\exists a_i \in A$ such that

$$\varphi^n + a_1 \varphi^{n-1} + \cdots + a_{n-1}\varphi + a_n = 0 \in End_A(M)$$

hence by the defnintion of φ, we get

$$(x^n + a_1 x^{n-1} + \cdots + a_n) m_i = 0 \ (i = 1, 2, \ldots, n)$$

which implies that

$$x^n + a_1 x^{n-1} + \cdots + a_n \in Ann(M) = \{0\}$$

thus $x \in B$ is integral over A. □

so we have the corollary

Corollary 0.1. *: Let $A \subseteq B$ be rings, if $b_1, b_2, \ldots, b_n \in B$ are all integral over A, then $A[b_1, b_2, \ldots, b_n]$ is a finitely generated A-module.*

Proof: By induction.

We have proved the case that $n = 1$;

Note that $A[b_1, b_2 \ldots, b_n] = (A[b_1, b_2, \ldots, b_{n-1}])[b_n]$. Since b_n is integral over A, so b_n is integral over $A[b_1, b_2, \ldots, b_{n-1}]$. So we can know that $A[b_1, b_2 \ldots, b_n] = (A[b_1, b_2, \ldots, b_{n-1}])[b_n]$ is a finitely generated $A[b_1, b_2, \ldots, b_{n-1}]$-module. By induction hypothesis, $A[b_1, b_2, \ldots, b_{n-1}]$ is a finitely generated A-module , thus $A[b_1, b_2 \ldots, b_n] = (A[b_1, b_2, \ldots, b_{n-1}])[b_n]$ is a finitely generted A-module. □

Corollary 0.2. *: Let $A \subseteq B$ be rings, suppose $C = \{x \in B | \ x \text{ is integral over } A\}$, then C is a subring of B containing A.*

Proof: Let $x, y \in C$, thus by Corollary 0.1, we know that $A[x, y]$ is a finitely generated A-module and $x + y, xy \in A[x, y]$. Hence $A[x + y], A[xy] \subseteq A[x, y]$ and therefore $A[x + y], A[xy]$ are both finitely generated A-module. By theorem 0.1, we know that $x + y, xy$ are both integral over A. □

Definition 0.2. : Let $A \subseteq B$ be rings, the set

$$C := \{x \in B |\ x \text{ is integral over } A\}$$

is called the **integral closure of A in B**; If $C = B$, we say B **is integral over** A; If $C = A$, we say A **is integral closed in** B.

Definition 0.3. : If A is an integral domain, then the integral closure of A is defined as the integral closure of A in its fraction field.

Example 0.5. : The integral closure of $\mathbb{Z}$ in $\mathbb{Q}$ is $\mathbb{Z}$.

Proof: For $\forall \frac{q}{p} \in \mathbb{Q}$, where $(p, q) = 1$, if $\frac{q}{p}$ is integral over $\mathbb{Z}$, then $\exists a_i \in \mathbb{Z}$ such that

$$\frac{q^n}{p^n} + a_1 \frac{q^{n-1}}{p^{n-1}} + \cdots + a_n = 0$$

which implies that $(p, q) \neq 1$ (a contradiction !) $\square$

Example 0.6. : If A is a UFD, then the integral closure of A is A itself.

Proposition 0.1. *: Let $A \subseteq B \subseteq C$ be rings, if C is integral over B and B is integral over A. Then C is integral over A.*

Proof: For $\forall x \in C$, $\exists b_1, b_2, \ldots, b_n \in B$ such that

$$x^n + b_1 x^{n-1} + \cdots + b_n = 0$$

Since $b_1, b_2, \ldots, b_n$ are all integral over A, by Corollary 0.1, we know that $A' := A[b_1, b_2, \ldots, b_n]$ is a finitely generated A-module. Since $x \in C$ is integral over $A' = A[b_1, b_2, \ldots, b_n]$, thus $A'[x]$ is a finitely gegerated A'-module, therefore $A'[x] = (A[b_1, b_2, \ldots, b_n])[x]$ is a finitely generated A-module, which implies that x is integral over A. $\square$

By this proposition, we have the following corollary

Corollary 0.3. *: If C is the integral closure of A in B, then C is integrally closed in B.*

Proposition 0.2. *: Let $A \subseteq B$ be rings, $J \subseteq B$ is an ideal of B, $I = J \cap A (= J^c)$ is the contracted ideal in A, if B is integral over A, then*
(1) B/J is integral over A/I;
(2) if S is a multipilcative set of A, then $S^{-1}B$ is integral over $S^{-1}A$.

Proof: (1)✓;
(2) For $\forall \frac{x}{s} \in S^{-1}B$, then $\exists a_i \in A$ such that

$$x^n + a_1 x^{n-1} + \cdots + a_{n-1} x + a_n = 0$$

which means that
$$(\frac{x}{s})^n + \frac{a_1}{s}(\frac{x}{s})^{n-1} + \cdots + \frac{a_{n-1}}{s^{n-1}}\frac{x}{s} + \frac{a_n}{s^n} = 0$$
where $\frac{a_i}{s^i} \in S^{-1}A$ and hence $S^{-1}B$ is integral over $S^{-1}A$. $\square$

Proposition 0.3. *: Let $A \subseteq B$ be integral extension of integral domains, i.e., A, B are both integral domains and B is integral over A. Then B is a field $\Leftrightarrow$ A is a field.*

Proof:($\Leftarrow$): For $\forall x \in B$ and $x \neq 0$, since x is integral over A, $\exists a_i \in A$ such that

$$x^n + a_1x^{n-1} + \cdots + a_n = 0$$

Since B is an integral domain, we may assume that $a_n \neq 0$ (or otherwise, if $a_n = 0$, then $x(x^{n-1} + a_1x^{n-2} + \cdots + a_{n-1}) = 0$, hence $x^{n-1} + a_1x^{n-2} + \cdots + a_{n-1} = 0$). Since A is a field and $a_n \neq 0$, so $a_n^{-1} \in A$ and hence $x^{-1} = -a_n^{-1}(x^{n-1} + a_1x^{n-2} + \cdots + a_{n-1}) \in B$, which means that B is a field;

($\Rightarrow$): For $\forall y \neq 0$ and $y \in A$, since B is a field, so $y^{-1} \in B$, hence y^{-1} is integral over A. Thus $\exists a_i \in A$ such that
$$(y^{-1})^m + a_1(y^{-1})^{m-1} + \cdots + a_{m-1}y^{-1} + a_m = 0$$
hence we get
$$1 + a_1y + \cdots + a_my^m = 0$$
which implies that
$$y^{-1} = -(a_1 + a_2y + \cdots + a_my^{m-1}) \in A$$
which shows that A is a field. $\square$

Corollary 0.4. *: Let $A \subseteq B$ be rings, B is integral over A. Let $\mathfrak{p} \subseteq B$ be a prime ideal of B and $\mathfrak{q} = \mathfrak{p} \cap A (= \mathfrak{p}^c)$. Then $\mathfrak{p}$ is a maximal ideal of B $\Leftrightarrow$ $\mathfrak{q}$ is a maximal ideal of A.*

Proof: By Proposition 0.2, we know that $B/\mathfrak{p}$ is integral over $A/\mathfrak{q}$ and $B/\mathfrak{p}, A/\mathfrak{q}$ are both integral domains. Thus we know that

$$\mathfrak{p} \text{ is maximal } \Leftrightarrow B/\mathfrak{p} \text{ is a field } \Leftrightarrow A/\mathfrak{q} \text{ is a field } \Leftrightarrow \mathfrak{q} \text{ is maximal. } \square$$